

Gear fault detection based on adaptive wavelet packet feature extraction and relevance vector machine

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Abstract: In this article, a novel fault detection method based on adaptive wavelet packet feature extraction and relevance vector machine (RVM) is proposed for incipient fault detection of gear. First, ten statistical characteristics in time domain and all node energies of full wavelet packet tree are extracted as candidate features. Then, Fisher criterion is applied to evaluate the discrimination power of each feature. Finally, two optimal features from time domain and wavelet domain, respectively, are selected to be used as inputs to the RVM. Furthermore, moving average is applied to each feature to improve accuracy for online continuous fault detection. By combining wavelet packet transform with Fisher criterion, it is able to adaptively find the optimal decomposition level and select the global optimal features. The RVM, a Bayesian learning framework of statistical pattern recognition, is adopted to train the fault detection model. The RVM was compared with the popular support vector machine (SVM) with the increase of training samples. Experimental results validate the effectiveness of the proposed method, and indicate that RVM is more suitable than SVM for online fault detection.

Keywords: condition monitoring, fault detection, fault diagnosis, relevance vector machine, support vector machine, wavelet packet transform

1 INTRODUCTION

Condition-monitoring system (CMS) plays a pivotal role in establishing a condition-based maintenance, which is considered to provide more benefits than the unplanned corrective maintenance and time-based preventive maintenance, through significantly reducing maintenance costs and improving equipment availability [1]. Autonomous online CMSs with integrated fault detection and diagnosis algorithms allow early warnings of mechanical and electrical defects to prevent major component failures. Many incipient faults can be detected while the defective component is still operational. Thus,

necessary repair actions can be planned in time and need not be taken immediately, and this fact is of special importance for mission critical equipments. Generally, automatic intelligence fault detection and diagnosis are treated as problems of pattern recognition, which mainly include three tasks: raw signal acquisition, feature extraction, and classification.

Feature extraction plays an important role in machinery fault detection. The classification performance depends heavily on the quality of the feature extraction. There is a need to make the classification process faster and accurate using the minimum number of features which primarily characterize the system conditions [2]. Fewer features need less memory and less processing time which is important for online application. Specially, if only two features

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are required, the results can be shown visually in a two-dimensional (2D) space, which is valuable for CMS. However, much more than two features were usually adopted in previous literatures. In order to keep high accuracy while using the number of features as few as possible, it is essential to extract the optimal features that provide the most discrimination information among the classes of vibration data. There are a number of feature extraction methods for vibration signals in the literature. A popular time–frequency domain method is applying wavelet transform (WT), which has obtained great success in machine fault diagnostics for its many distinct advantages [3–9]. However, WT cannot split the high-frequency band where the modulation information of machine fault always exists [8]. Thus, the wavelet package transform (WPT), which contains complete decomposition at every level, can overcome the difficulty. The WPT was widely used as a feature extraction tool for machine fault detection and diagnosis [6–9]. The wavelet packet node energy representation could provide a more robust signal feature for classification than using the wavelet packet coefficients directly [6]. In previous literatures, node energies of the bottom level in wavelet packet tree (WPT) were commonly taken as features for classification. The number of decomposition level was usually chosen simply by rule of thumb. For example, Yen and Lin [6] chose 7-level, Chen *et al.* [7] 5-level, and Hu *et al.* [8] and Huang *et al.* [9] 3-level. In such cases, they might fail to extract the optimal features which might be hidden in some other levels. Therefore, it is important to choose a proper decomposition level. To address this problem, the well-known ‘best basis’ algorithm developed by Coifman and Wickerhauser [10] and its upgrade versions were usually used [11–13]. They used an entropy-based criterion to select the most suitable decomposition of a given signal to find a ‘best tree’ which can represent the signal in the most compact way. These algorithms were developed originally for signal compression. They are not best suitable for feature detection in classification, because, in contrast to compression which removes redundant information but maintains all signal features, classification only cares about those signal features that contain discrimination information among different classes. In this study, an adaptive wavelet feature extraction approach is proposed to address the above-mentioned problems. It can find the optimal decomposition level of WPT from classification point of view, and obtain the global optimal features adaptively from node energies of full WPT by combining WPT with Fisher criterion.

Recently, various intelligent classification algorithms have been successfully applied to automated detection and diagnosis of machine conditions. Artificial neural networks (ANN) [7, 11, 14, 15] and support vector machines (SVM) [2, 8, 9, 16] are used the most widely. The main difference between ANN and SVM is in their risk minimization. In the case of SVM, structural risk minimization principle is used to minimize an upper bound based on an expected risk. In the case of ANN, traditional empirical risk minimization is used to minimize the error in training of data. The difference in risk minimization leads to a better generalization performance for SVM than ANN [8]. The SVM has become widely established as one of the leading approaches to pattern recognition and machine learning, combining excellent generalization properties with sparse kernel representation. However, despite its widespread success, SVM suffers from some significant practical limitations [17]: (1) the number of support vectors required typically grows linearly with the size of the training set. This increases computational complexity; (2) it makes a ‘hard’ binary decision with no probabilistic which is ideally of great use for reasonable decision of maintenance. The relevance vector machine (RVM), originally introduced by Tipping [17] in 2001, is a Bayesian treatment and does not suffer from any of the two limitations. The most compelling feature of the RVM is that, it achieves comparable classification accuracy to the SVM, provides a full predictive distribution, and requires dramatically fewer kernel functions so that it consumes much less test time which is important in practice, such as online fault detection. Recently, RVM has been used successfully for the solution of many problems including object detection [18], face tracking [19], model identification [20], cancer diagnosis [21], image classification [22], and so on. Although the RVM has been claimed to have remarkable properties, attentions paid to the RVM are far less than the SVM. The applications of the RVM in the literatures of machine condition monitoring and diagnosis are very rare. Lei and Zhang [23] had applied the RVM for bearing fault detection and diagnosis. Widodo *et al.* [24] had used RVM and SVM for fault diagnosis of low-speed bearing based on acoustic emission and vibration signals.

In this study, a novel fault detection method based on adaptive wavelet packet feature extraction and RVM is proposed for incipient fault detection of gear. The vibration signals were measured at the transmission of a pump with normal and incipient defective gears under different load conditions, in a pump station at the Netherlands [25]. Two features that extracted from time domain and wavelet domain, respectively, were finally used to make up a

2D feature space for classification. The results demonstrate the effectiveness of the proposed method.

The rest of this article is organized as follows. Section 2 provides the adaptively wavelet packet feature extraction methodology with related mathematics. Section 3 elaborates the principle of the RVM for classification. In Section 4, experiment and discussion are given. Section 5 provides some conclusions.

2 ADAPTIVE WAVELET PACKET FEATURE EXTRACTION

2.1 Overview

In most previous studies, node energies of the bottom level in WPT were commonly taken as features for classification. In order to obtain the optimal features, it is essential to select the optimal decomposition level. The widely used entropy-based algorithms were designed to select the most suitable decomposition level from the compression point of view. It might be unable to find the best level for classification purpose. In this article, this problem is addressed from the classification point of view directly. First, the global optimal features are searched from the full WPT with high-decomposition level. Then, adaptively, the optimal decomposition level is found. The optimal features are those providing the most discrimination information among different classes. They are finally used as inputs to classifier while the other features that provide little information useful to classification purpose are abandoned.

The application procedure of the adaptive wavelet packet feature extraction mainly includes five steps as follows.

Step 1: Compute the full wavelet packet decomposition (WPD) on each training sample for L resolution levels, where L should be very high, e.g. six or seven.

Step 2: Take all node energies in all levels of the full WPT as candidate features, and normalize them.

Step 3: Evaluate the discrimination power of each feature with respect to a certain criterion. Then, the optimal features are selected from the candidate features that have larger criterion values. Fisher criterion was used in this study.

Step 4: Take the node positions of these optimal features as the optimal nodes. The optimal decomposition level is obtained by finding the maximum level of the optimal nodes, and the best WPT, which is the smallest binary tree that contains all the optimal nodes, is also worked out.

Step 5: For each testing sample, compute only the best WPT rather than the full tree as in step 1;

node energies of the optimal nodes are then calculated to form the feature vector.

2.2 Wavelet packet transform and node energy

Wavelet analysis decomposes a signal into two parts, approximations and details. Approximations are the high-scale low-frequency components while details are the low-scale high-frequency components. Approximations give global information of a signal while details give detailed information of a hidden pattern in the signal. The wavelet packet transform is a generalization of wavelet decomposition that offers a richer range of possibilities for signal analysis. In wavelet packet analysis, the details as well as the approximations can be split. This yields more than different ways to encode the signal [10, 26].

The wavelet functions ψ^j can be obtained from the following recursive relationships

$$\psi^{2j}(t) = \sqrt{2} \sum_{k=-\infty}^{\infty} h(k) \psi^j(2t - k) \quad (1)$$

$$\psi^{2j+1}(t) = \sqrt{2} \sum_{k=-\infty}^{\infty} g(k) \psi^j(2t - k) \quad (2)$$

The first wavelet is so-called a mother wavelet function as follows

$$\psi^1(t) = \psi(t) \quad (3)$$

The discrete filters $h(k)$ and $g(k)$ are the quadrature mirror filters associated with the scaling function and the mother wavelet function. There are quite a few mother wavelets reported in the literature. Daubechies [26] developed a family of mother wavelets based on the solution of a dilation equation. One of these wavelet functions, DB5, was adopted in this study.

A wavelet packet $\psi_{j,k}^n(t)$ is a function with three indices where integers n , j , and k are the oscillation, scale, and translation parameters, respectively.

$$\psi_{j,k}^n(t) = 2^{j/2} \psi^n(2^j t - k), \quad n = 1, 2, 3 \dots \quad (4)$$

The WT consists of one high-frequency term from each level and one low-frequency residual from the last level of decomposition. The WPT, on the other hand, contains a complete decomposition at every level and hence can achieve a higher resolution in the high-frequency region. After j level of decomposition, the original signal $f(t)$ can be expressed as

$$f(t) = \sum_{n=1}^{2^j} f_j^n(t) \quad (5)$$

The wavelet packet component signal $f_j^n(t)$ can be represented by a linear combination of wavelet packet functions $\psi_{j,k}^n(t)$ as follows

$$f_j^n(t) = \sum_{k=-\infty}^{\infty} \omega_{j,k}^n \psi_{j,k}^n(t) \quad (6)$$

where the wavelet packet coefficients $\omega_{j,k}^n$ can be obtained from

$$\omega_{j,k}^n = \int_{-\infty}^{\infty} f(t) \psi_{j,k}^n(t) dt \quad (7)$$

Direct use of the wavelet packet coefficients as features often lead to difficult and inaccurate classification. An alternative is the wavelet packet node energy, which is defined as follows

$$E_{j,n} = \sum_k \left(\omega_{j,k}^n \right)^2 \quad (8)$$

which measures the signal energy in a specific frequency band indexed by the scale parameter j and the oscillation parameter n .

Each wavelet packet node energy value can be used as an individual feature. Suppose that, computing the full WPD on a signal segment for L resolution levels, the result is a group of $2^1 + 2^2 + \dots + 2^L = 2^{L+1} - 2$ sets of coefficients where each set corresponds to a wavelet packet node. Hence, $(2^{L+1} - 2)$ node energies can be obtained and a $(2^{L+1} - 2)$ dimensional feature vector formed.

2.3 Fisher criterion

In a two-class problem, given a set of samples $\{\mathbf{x}_i\}_{i=1}^N$, where $\mathbf{x}_i = \{\mathbf{x}_{i,j}\}_{j=1}^D$ and N is the number of samples, in which N_1 samples belong to class C_1 and N_2 samples belong to class C_2 and each sample $\mathbf{x}_i$ is a D -dimension feature vector. The Fisher criterion for one single feature is defined to be the ratio of the between-class variance to the within-class variance and given by

$$J(f_j) = \frac{S_{B,j}}{S_{W,j}} \quad j = 1, 2, \dots, D \quad (9)$$

where f_j is the j th feature, $S_{B,j}$ the between-class variance of f_j , and $S_{W,j}$ the total within-class variance of f_j , given by

$$S_{B,j} = (m_{1,j} - m_{2,j})^2 \quad (10)$$

$$S_{W,j} = \sigma_{1,j}^2 + \sigma_{2,j}^2 \quad (11)$$

where $m_{1,j}$ is the mean of f_j in class C_1 , $m_{2,j}$ the mean of f_j in class C_2 , $\sigma_{1,j}^2$ the within-class variance of f_j in

class C_1 , and $\sigma_{2,j}^2$ the within-class variance of f_j in class C_2 , given by

$$m_{1,j} = \frac{1}{N_1} \sum_{i \in C_1} \mathbf{x}_{i,j} \quad (12)$$

$$m_{2,j} = \frac{1}{N_2} \sum_{i \in C_2} \mathbf{x}_{i,j} \quad (13)$$

$$\sigma_{1,j}^2 = \frac{1}{N_1 - 1} \sum_{i \in C_1} (\mathbf{x}_{i,j} - m_{1,j})^2 \quad (14)$$

$$\sigma_{2,j}^2 = \frac{1}{N_2 - 1} \sum_{i \in C_2} (\mathbf{x}_{i,j} - m_{1,j})^2 \quad (15)$$

With respect to Fisher criterion, the feature with larger Fisher criterion value can give a wider margin of separation between classes, meanwhile giving smaller variance within class. Thereby, there is less class overlap which gives more discrimination power and can be used for feature selection for classification.

3 RELEVANCE VECTOR MACHINE FOR CLASSIFICATION

The relevance vector machine proposed by Tripping [17] is a model of identical functional form to the popular SVM. The learning of RVM is under the Bayesian framework, and it yields a probabilistic output.

3.1 Model specification

Given a set of input-target training pairs $\{\mathbf{x}_i, t_i\}_{i=1}^N$, where $\mathbf{x}_i \in R^n$, $t_i \in \{0, 1\}$ and N is the number of samples. The RVM makes predictions for new inputs $\hat{\mathbf{x}}$ based on the SVM-like function; the model takes the form of a linear combination of basis functions transformed by a logistic sigmoid function

$$y(\hat{\mathbf{x}}, \mathbf{w}) = \sigma \left(\sum_{i=1}^N \omega_i k(\mathbf{x}_i, \hat{\mathbf{x}}) + \omega_0 \right) = \sigma(\mathbf{w}^T \mathbf{K}) \quad (16)$$

where $\mathbf{w} = (\omega_0 \dots \omega_N)^T$ is the weight vector, $\mathbf{K}(\mathbf{x}) = [1, k(\mathbf{x}_1, \mathbf{x}) \dots k(\mathbf{x}_N, \mathbf{x})]^T$ the kernel function vector, and $\sigma(\cdot)$ the logistic sigmoid function defined by

$$\sigma(a) = \frac{1}{1 + \exp(-a)} \quad (17)$$

The logistic sigmoid function plays an important role in many classification algorithms; it satisfies the following symmetry property

$$\sigma(-a) = 1 - \sigma(a) \quad (18)$$

Therefore, model (16) can be used as the posterior probability of the membership of one of the classes

given the input x . The posterior probability of class C_1 can be written as

$$P(t = 1|x) = y(x, w) \quad (19)$$

Similarly, for class C_2 , it is seen that

$$P(t = 0|x) = 1 - y(x, w) \quad (20)$$

Due to the assumption of independence of the t_i , by adopting the Bernoulli distribution for $P(t|x)$, the likelihood of the complete dataset can be written as

$$P(t|w) = \prod_{i=1}^N [y(x_i, w)]^{t_i} [1 - y(x_i, w)]^{1-t_i} \quad (21)$$

where $\mathbf{t} = (t_1, \dots, t_N)^T$.

To avoid the over-fitting, RVM adopts a Bayesian perspective, and the zero-mean Gaussian prior distribution is used to assign an individual hyperparameter α_i to each weight ω_i . This is the key feature of the RVM. Thus, the weight prior distribution takes the form

$$p(w|\alpha) = \prod_{i=0}^N \mathcal{N}(\omega_i | 0, \alpha_i^{-1}) = \left(\frac{\alpha}{2\pi}\right)^{(N+1)/2} \exp\left\{-\frac{\alpha}{2} w^T w\right\} \quad (22)$$

where α_i represents the precision of the corresponding parameter ω_i , and $\alpha = (\alpha_0, \dots, \alpha_N)^T$.

3.2 Optimizing the weights and hyperparameters

For the current, fixed values of α , the ‘maximum a posteriori’ (MAP) estimate for w is the single most probable value under the posterior distribution. Now, given the likelihood and the prior, the posterior distribution over w via Bayes’ rule is computed

$$p(w|t, \alpha) = \frac{P(t|w)p(w|\alpha)}{p(t|\alpha)} \quad (23)$$

Since the denominator in (23) is independent of w , MAP is equivalent to the maximum of the numerator, that is $p(w|t, \alpha) \propto P(t|w)p(w|\alpha)$. Retaining only those terms dependent on w gives

$$\begin{aligned} w_{\text{MP}} &= \arg \max_w P(t|w)p(w|\alpha) \\ &= \arg \max_w \log\{P(t|w)p(w|\alpha)\} \\ &= \arg \max_w \left\{ \sum_{i=1}^N [t_i \log y_i + (1 - t_i) \log(1 - y_i)] \right. \\ &\quad \left. - \frac{1}{2} w^T A w \right\} \end{aligned} \quad (24)$$

where $A = \text{diag}(\alpha_0, \alpha_1, \dots, \alpha_N)$, $y_i = y(x_i, w)$.

The likelihood $P(t|w)$ is not a Gaussian distribution; so, the MAP estimation w_{MP} cannot be computed analytically. Thus, the Laplace approximation is utilized. The first step is initializing the hyperparameter vector α . For this given value of α , a Gaussian approximation to the posterior distribution is then built, and thereby an approximation to the marginal likelihood is obtained. Maximization of this approximate marginal likelihood then leads to a re-estimated value for α , and the process is repeated until convergence. See references [17, 27] for more details. In practice, during re-estimation, many of the hyperparameters are driven to large values, so that the corresponding weights are effectively force to zero. Thus, the corresponding basis functions can be pruned, and sparsity is realized. The inputs x_i corresponding to the remaining non-zero weights are called *relevance vectors* (RVs).

3.3 Probabilistic estimation

For a new input vector $\hat{x}$, the probabilistic discriminative model (16) can be rewritten only based on the w_{MP} and RVs as follows

$$y(\hat{x}, w_{\text{MP}}) = \sigma\left(\sum_{x_i \in \text{RVs}} \omega_i k(x_i, \hat{x}) + \omega_0\right) \quad (25)$$

Then, the posterior probabilities of both classes can be obtained by equations (19) and (20), respectively. The classification boundary, corresponding to the minimum misclassification rate, occurs at where the two posterior probabilities are equal.

After training, the RVM makes prediction only requiring a small amount of computation based on a few RVs. Hence, it can be implemented for online applications. Furthermore, the full predicted probability of class membership provides more rational evidence for decision making. These features make RVM very attractive and suitable for online machinery fault detection and diagnosis.

4 EXPERIMENT AND DISCUSSION

4.1 Experimental setup

The vibration data used in this article came from the public resources by Delft University of Technology [25]. The machine under investigation was a pump driven by an electromotor. The incoming shaft was reduced in speed by two decelerating gear-combinations. Measurements were repeated for two identical machines, where the first one showed a progressed pitting in both gears (a common incipient fault of gears), and the second one was virtually fault free (normal). Data from sensor 1, which was mounted

near the driving shaft, were used in our research. Four sets of measurements were obtained separately from the two machines under two levels of load conditions: minimum and maximum loads. For each condition, data were measured for 24.3 s, low pass-filtered to 1 kHz, and digitized at 3.2 kHz. Both load states have been included in order to obtain a flexible diagnosis towards changes in the load conditions. Samples of the raw vibration signals of different conditions were plotted in Fig. 1.

4.2 Feature extraction

In order to extract features which were sensitive to different gear conditions but insensitive to varying load conditions, all four datasets were combined into two sets by combining datasets from the same machine under different load conditions. Then, the original datasets were divided into 600 segments of 1024 points each. Each segment was used as one sample; hence, 300 samples of normal and 300 samples of abnormal (pitting) were obtained.

For each sample, six-level WPD was generated. A six-level WPT was plotted in Fig. 2(a). Then, all node energies of the full WPT were computed and formed 126 features (i.e. $2^{6+1} - 2 = 126$). Meanwhile, ten statistical characteristics in time domain (i.e. mean, maximum amplitude, root mean square, peak-to-peak

value, pulse index, crest-factor, waveform factor, standard variance, skewness, and kurtosis) were also extracted. Therefore, 136 features were obtained for each sample. In order to avoid features in greater numeric ranges dominate those in smaller numeric ranges, all features were normalized linearly to the range of [0, 1].

Finally, Fisher criterion was employed to evaluate the discrimination power of each feature for feature selection. The results were displayed in Fig. 3. It can be seen that, the Fisher criterion values of two features are significantly greater than those of the others. The first one is the maximum amplitude in time domain. The second one is the node energy of node 27 at level 4 in wavelet domain. Therefore, these two features were selected as the optimal features to construct a 2D feature space for fault detection.

Among wavelet domain features, as can be seen in Fig. 3, in this case, the optimal feature was hidden at level 4. If each sample was decomposed down to any other level except level 4, and node energies of the bottom level were used only as most previous works, we would fail to obtain suitable features for classification. Hence, one advantage of this proposed method is that it can obtain the optimal features adaptively. In this study, only one optimal feature in wavelet domain was selected. However, of course, for other situations, this method could be used to select

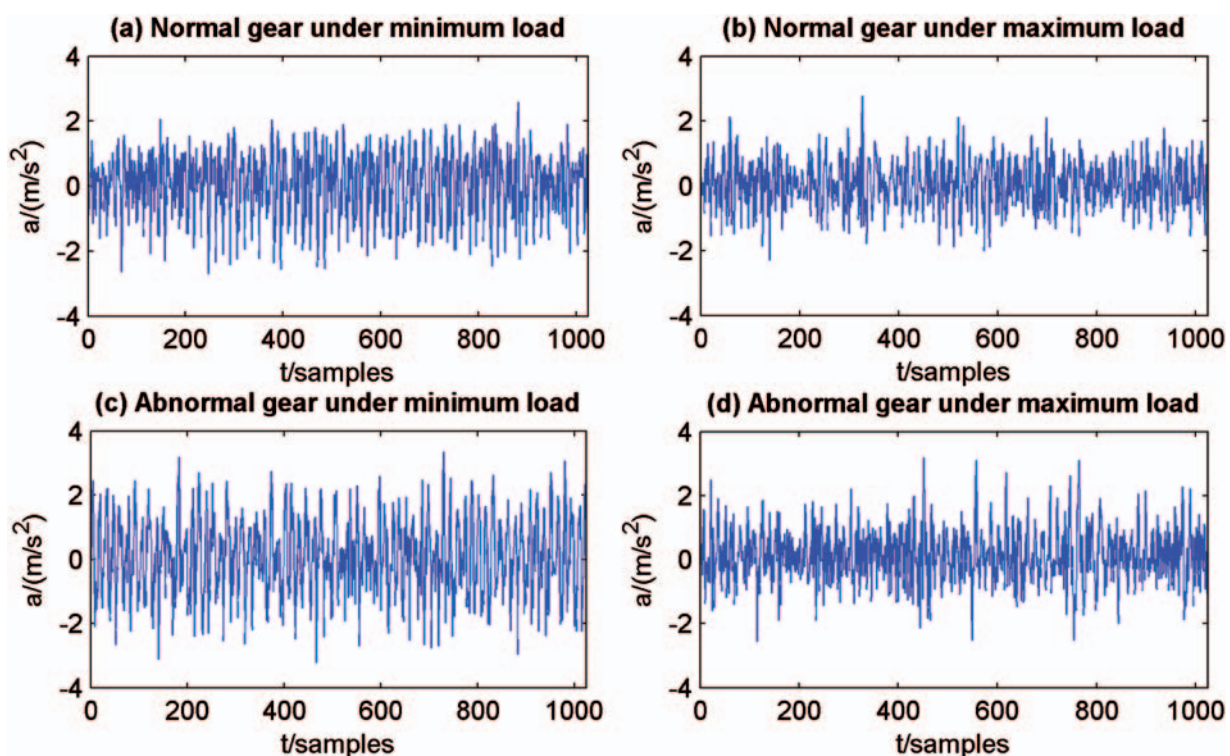


Fig. 1 Vibration signals of gears in different conditions

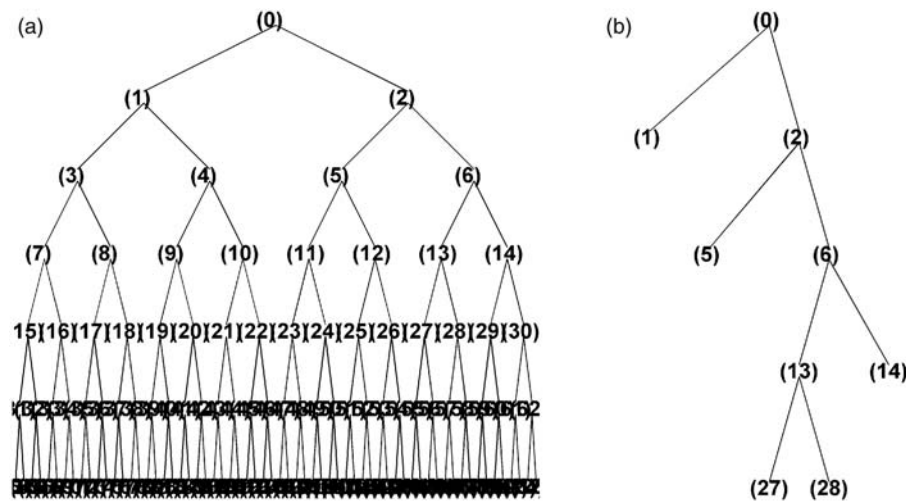


Fig. 2 (a) Full wavelet packet tree at level 6. (b) The best wavelet packet tree

out more than one optimal feature that have the highest Fisher criterion values. Also, these features might come from different levels.

After the optimal feature is obtained, as given in Section 2.1, node 27 is the optimal node, and level 4 the optimal decomposition level. The best WPT is consequently achieved, as shown in Fig. 2(b). The results show that the best WPT is much smaller than the initial tree, as shown in Fig. 2(a). In practice, only the best WPT and the optimal feature need be calculated, which implies significant simplification of the WPD and dramatic saving of computing time and memories.

4.3 Fault detection based on RVM

4.3.1 Comparison of RVM and SVM. In order to evaluate the proposed classification method, both RVM and SVM classification methods have been applied for fault detection upon the 2D feature space obtained after feature extraction. In both cases, the Gaussian radial basis function kernel was used, and the fivefold cross-validation applied to tune the optimal kernel parameter. There were in total 600 samples; 300 samples of normal and 300 samples of abnormal (pitting). They were divided into training and testing samples through a random process; a

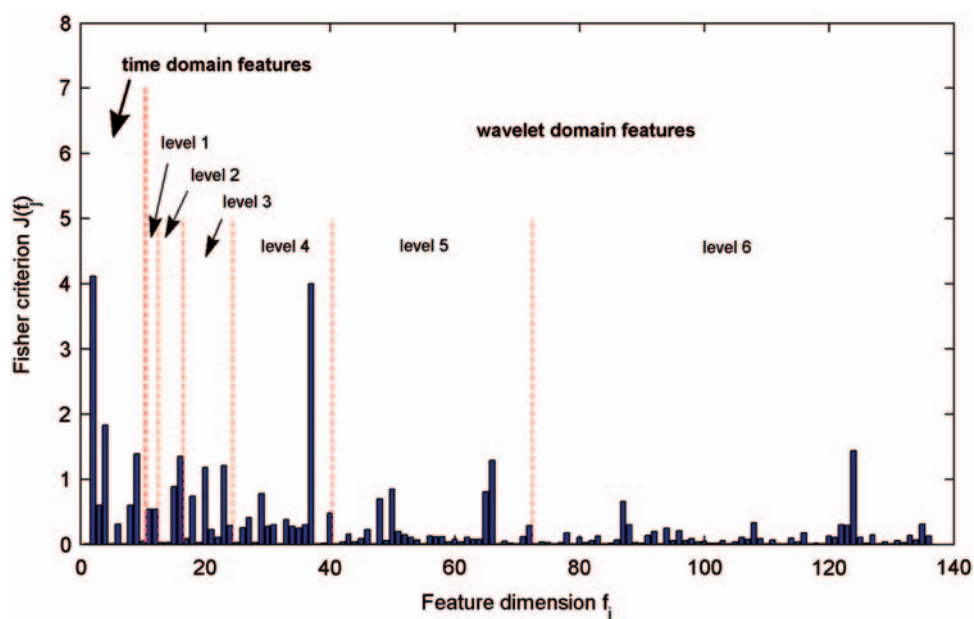


Fig. 3 Fisher criterion of features

Table 1 Performance comparisons of RVM and SVM with increase of training samples

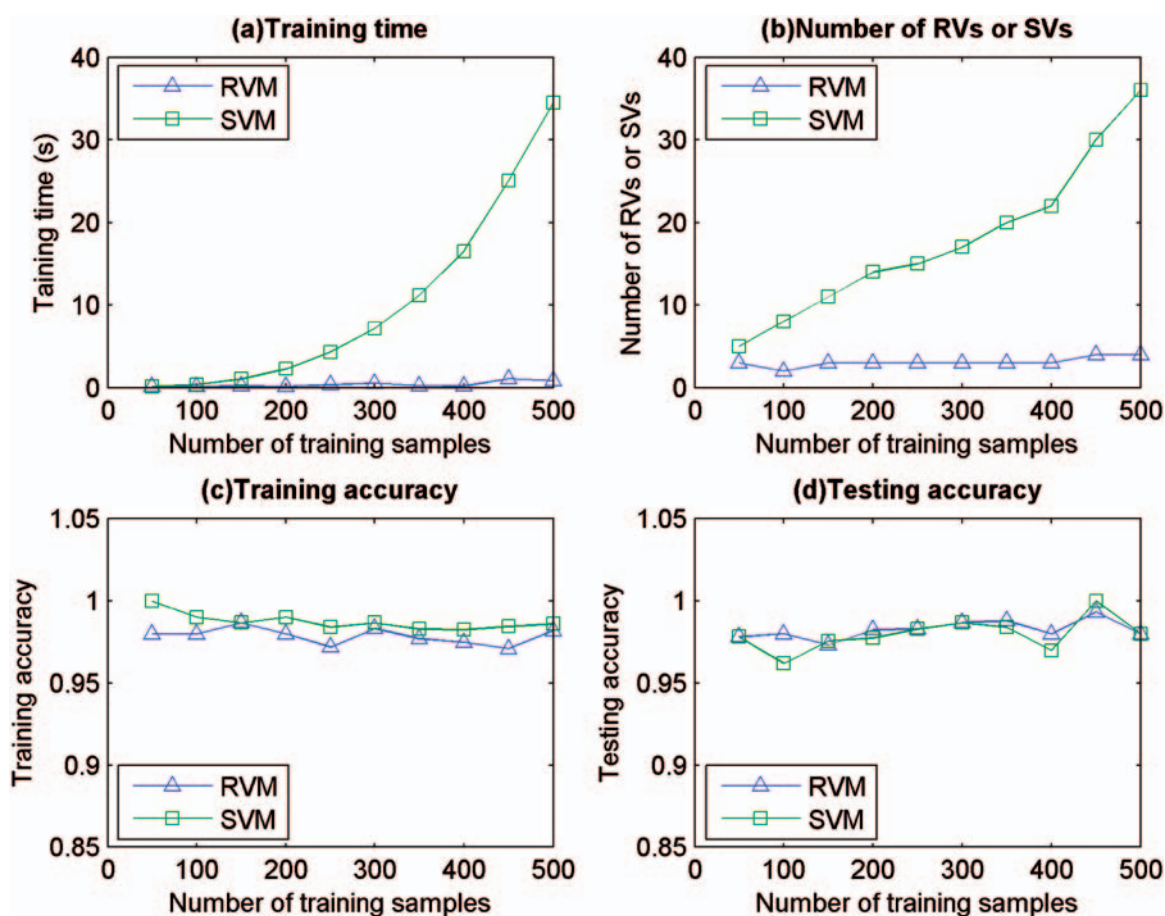
Tr.No	Te.No	RVM				SVM			
		Tr.Time (s)	RV.No	Tr.Ac	Te.Ac	Tr.Time (s)	SV.No	Tr.Ac	Te.Ac
50	550	0.0781	3	0.9800	0.9782	0.0781	5	1	0.9782
100	500	0.0781	2	0.9800	0.9800	0.3594	8	0.99	0.9620
150	450	0.1718	3	0.9867	0.9733	1.0313	11	0.9867	0.9757
200	400	0.1250	3	0.9800	0.9825	2.2656	14	0.99	0.9775
250	350	0.3125	3	0.9720	0.9829	4.2969	15	0.984	0.9829
300	300	0.5312	3	0.9833	0.9867	7.1563	17	0.9867	0.9867
350	250	0.2187	3	0.9771	0.9880	11.172	20	0.9829	0.9840
400	200	0.1875	3	0.975	0.9800	16.516	22	0.9825	0.9700
450	150	1.0469	4	0.9711	0.9933	25.031	30	0.9844	1.000
500	100	0.8125	4	0.9820	0.9800	34.453	36	0.9860	0.9800

certain number of samples were used for training purpose and the rest for testing.

In this study, the performance of RVM and SVM with the increase of training samples are compared. Experimental results are listed in Table 1, and meanwhile, plotted in Fig. 4. Note that in Table 1, 'Tr.No' refers to the number of training samples, 'Te.No' the number of testing samples, 'Tr.Time' the training time, 'Tr.Ac' the training accuracy, 'Te.Ac' the testing

accuracy, 'RV.No' the number of RVs, and 'SV.No' the number of support vectors (SVs).

It is seen that with the increase of the training samples, both RVM and SVM can always achieve high and comparable classification accuracy. Specifically, the training accuracies of RVM are generally equal or a little lower than that of SVM. However, the testing accuracies of RVM are generally equal or a little higher than that of SVM. The results imply that

**Fig. 4** Performance comparisons of RVM and SVM with increase of training samples

RVM has better generalization performance to unseen samples than SVM. Moreover, the results also show that RVM has good generalization ability over small training dataset.

With the increase of training samples, when SVM is adopted, the training time and the number of SVs increase remarkably. However, when RVM is adopted, the training time and the number of RVs show very little fluctuation. Specially, when training dataset is large, in contrast to SVM, the training process of RVM model consumes much less time, and the number of RVs is significantly fewer (i.e. 4 *versus* 36). Since RVM and SVM have identical functional forms based on a linear combination of kernel functions, the computational complexity of the decision function is linearly proportional to the number of RVs or SVs. The RVM with dramatically fewer RVs implies a significant reduction in computational complexity of the decision function, thereby making it more suitable in real application. Furthermore, from the algorithm point of view, it implies a better ability to generalize on unseen data points when using fewer RVs [21]. Also, this has been validated as mentioned above.

In practice, using more training samples can generally lead to better generalization ability, since with the more training samples, the samples' distribution can generally be closer to the real distribution. As the application of CMS goes by, more samples will be available and can be used to upgrade the old model. With the rapid increase of training samples, the SVM model will become very complicated or will have to

throw out some useful samples. From this point of view, RVM has better upgradeability than SVM.

In order to further investigate the characteristics of the RVM model, the experimental results of the use of 300 training and 300 testing samples are plotted in the 2D feature space, as shown in Figs 5 and 6. It is seen that the boundaries of the two methods are similar, but the RVM only utilizes three RVs to construct the boundary line while almost all the input vectors near the boundary are chosen as SVs for the SVM.

Moreover, as can be seen in Fig. 5, the RVM model provides a full posterior probability distribution (PPD) of class membership rather than make a simple 'hard' binary decision as the SVM. The red line is the classification boundary where the posterior probability equals 50 percent. The gradient colour from yellow to green represents the PPD where all the points on each of the contour lines have the same posterior probability as shown by text (e.g. 0.8, 0.9). Only the posterior probability of the abnormal class is presented as given by equation (19), and the posterior probability of the normal class can be simply obtained through equation (20). The 2D feature space with PPD can be very useful since it can provide a visual representation of machine conditions and help the CMS make more rational decisions.

4.3.2 Fault detection with feature moving average. In the above experiment, each testing sample was treated individually to evaluate the status of the machine, which has achieved some

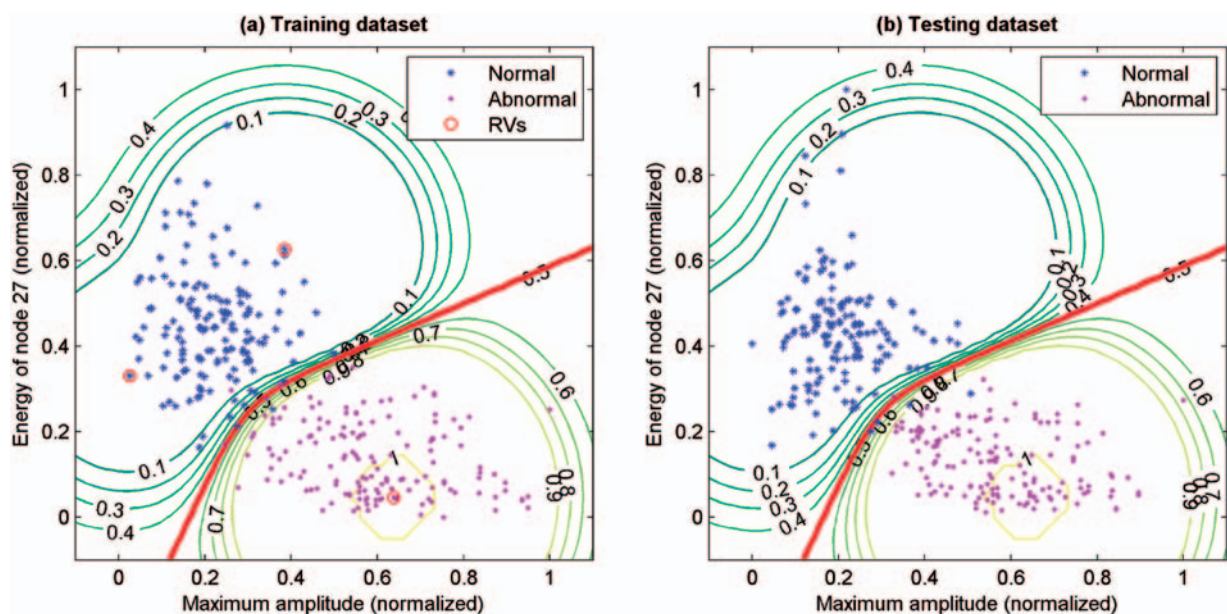


Fig. 5 Training and testing results of the RVM model

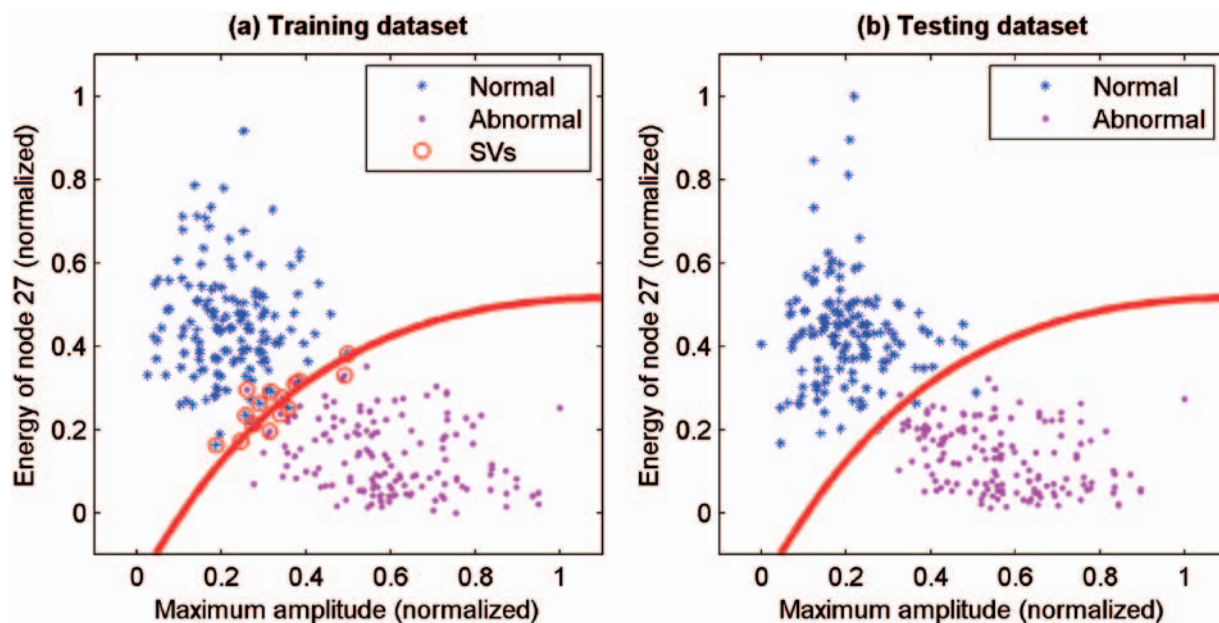


Fig. 6 Vibration signals of gears in different conditions

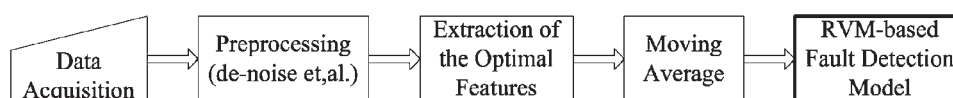


Fig. 7 Flowchart of on-line fault detection

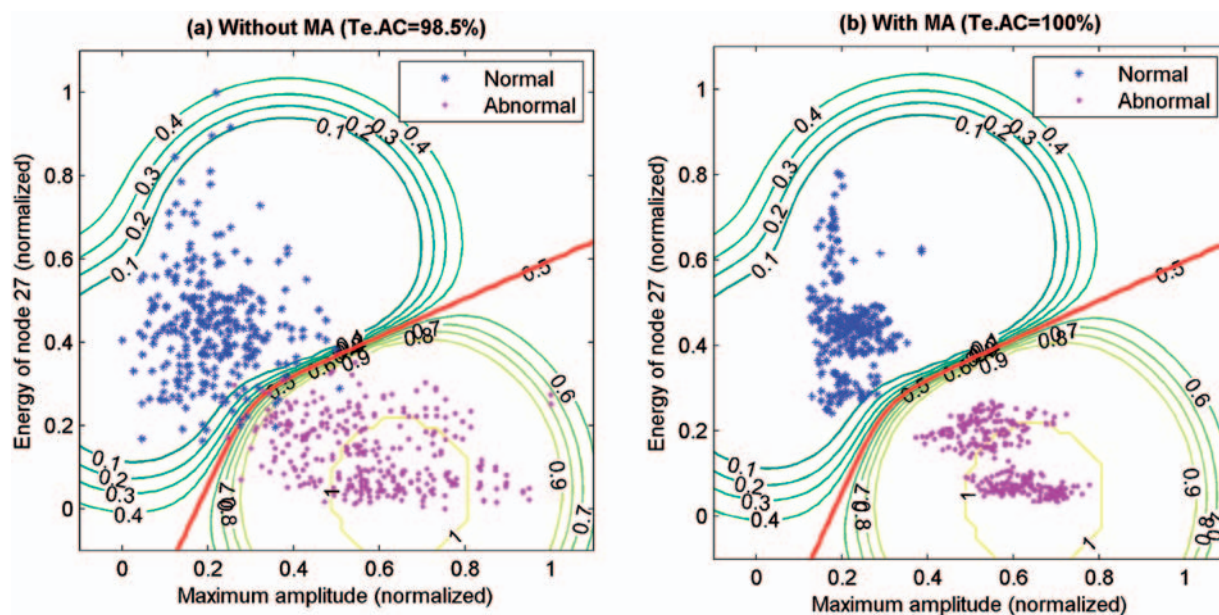


Fig. 8 Testing results of the RVM model with and without moving average

wrong decision results. In real application of CMS, segments of vibration signals are acquired continuously; the optimal features are then extracted from each segment. Along the time line, each feature can

be treated as one data series. Therefore, in order to smooth out short-term fluctuations and improve fault detection accuracy, moving average (MA) can be applied to each feature before it is input into the

fault detection model. The flowchart of the proposed online fault detection method is shown in Fig. 7.

The MA model of one feature is defined by

$$\bar{x}_n = \frac{1}{L}(x_n + x_{n-1} + \cdots + x_{n-L+1}) \quad (26)$$

where L is the size of moving window, x_n the value of the feature of the n th segment, and $\bar{x}_n$ the MA which replaces x_n to be used as input to the fault detection model.

In order to simulate real application and validate the advantage of this proposed method, each of the original four datasets (Section 4.1) was divided into 150 segments of 1024 points in order, and then the two optimal features were extracted and normalized, resulting in four testing datasets of 150 2D samples each. The RVM-based fault detection model used for this testing was the model obtained by training over 300 training samples in Section 4.3.1. For comparison purpose, the model was tested in two ways. First, all the 600 testing samples were tested individually without MA; the result is shown in Fig. 8(a). Second, the four testing datasets were tested separately. For each of the above testing datasets, all samples were inputted into the RVM-based fault detection model one by one and MA was applied to each feature, in which the size of moving window was set to 8. The distribution of all the samples after MA and the testing result is shown in Fig. 8(b). It is seen that MA has made all samples gathered and the overlap disappeared, resulting in testing accuracy of 100 per cent. Experimental results demonstrate that by applying MA to each feature, it is able to significantly improve the fault detection accuracy. The experimental results also further validate the effectiveness of the optimal features and the RVM-based fault detection model.

5 CONCLUSION

In this article, a novel method for gear incipient fault detection based on WPD and RVM has been presented. An adaptive wavelet packet feature extraction approach was proposed to extract the optimal features from node energies of full WPT by combining WPT with Fisher criterion. One advantage of the proposed approach is that it can obtain the global optimal features while adaptively find the optimal decomposition level, and achieve the best WPT. Two optimal features with the most discrimination power from time domain and wavelet domain, respectively, were selected out and formed a 2D feature space for fault detection. The RVM model trained on these features has performed with high-classification accuracy, which validated the effectiveness of the

proposed feature extraction methods. Furthermore, according to the characteristics of continuous data acquisition and analysis in real application, MA has been applied to each feature before it is input into the fault detection model, which has improved the fault detection accuracy significantly to 100 per cent.

Both RVM and SVM classification methods have been applied for fault detection with the increase of training samples. Experimental results demonstrate that in contrast to SVM, RVM has better generalization performance to unseen samples, consumes much less training time, has much simpler decision function, and has better upgradeability. In addition, the RVM model provides more information than the SVM for uncertainty by offering full PPD rather than output 'hard' binary decision. The results indicate that the RVM is more suitable than the SVM for online application. It is effective and has powerful potential for fault detection and diagnosis of CMS.

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